

From Data Deluge to Data Strategy: A Practical Playbook for Music in the Age of AI

Part 2: Data Strategy. A literature-grounded expansion of "From Data Deluge to Data Strategy: Get the Power of Insights," presented by Isabella Pighi and Chiara Santoro at [SXSW 2025](#) (Music and Tech track), and the companion to Part 1 ("The Data Deluge"). Updated to mid-2026.

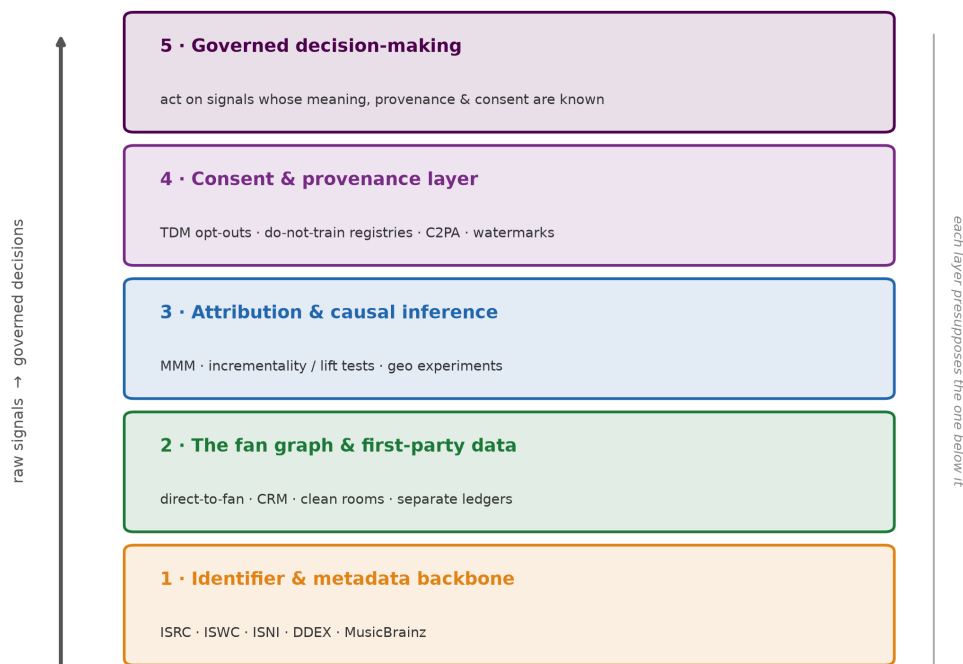
Abstract

Part 1 laid out the problem. The music industry is drowning in data that is scattered, skewed, warped by its own algorithms, and impossible to compare cleanly from one platform to the next. The fans who matter most spend most of their love in places no metric counts well. And AI has started flooding streaming libraries with fake tracks while helping fraudsters game the payout system. This second part is about what to actually do. Its core claim is simple: a data *strategy*, as opposed to a data deluge, is a stack, and you only get value at the top if every layer underneath is solid. There are five layers. At the bottom sits a backbone of standard identifiers and metadata (the codes that say which recording, which song, and which person a data point refers to), because without it nothing can be matched to anything else, and today hundreds of millions of dollars a year sit unpaid in "black boxes" for exactly this reason. Above that comes a fan graph built on data you own, so you can finally see the superfans Part 1 said were driving the business. The third layer swaps the "things happening together look like cause and effect" mistake for the discipline that fixes it: real experiments, incrementality tests, and geo tests, backed by twenty years of evidence that guessing from observation alone leads you astray. The fourth layer lets different companies combine their data without exposing anyone's personal information, through the privacy technology now sold as "data clean rooms," all under the rules of the GDPR. The top layer governs what feeds the AI: consent, opt-outs, "do-not-train" registries, content provenance, and watermarking, which are shifting from voluntary good manners to legal requirements under the EU AI Act. One idea ties the whole thing together: know what a signal means, and know whether you are allowed to use it, before you act on it.

1. Strategy is a stack, not a dashboard

Ask most people what a data strategy looks like and they will describe a nicer dashboard. That instinct is exactly what Part 1 warned against. A dashboard shows you numbers. It does nothing to make those numbers trustworthy, comparable, or safe to bet on. The argument of this second part is that the work which *does* make numbers trustworthy is structural, it comes in layers, and the layers have to be built in order. You cannot honestly credit a marketing campaign for a streaming spike if you cannot even reliably identify the recording. You cannot instrument a superfan relationship if you do not own the data that lets you see it. Figure 1 draws this as a stack, with an identifier and metadata backbone at the bottom, a fan graph and a proper measurement layer in the middle, and a consent-and-provenance layer at the top governing what reaches the AI models. Volume enters at the bottom. Value only comes out the top, and only if every layer beneath it holds.

The data-strategy stack: from a metadata backbone to governed decisions



Part 2's playbook. Volume enters at the bottom; value is only realised at the top, and only if every layer beneath holds.

This picture also answers a question Part 1 left hanging. Part 1 showed that the industry's attempts to make numbers comparable had all been local and arbitrary, with proprietary indices squashing a many-sided performance into one arguable score, and it demanded discipline without saying how. The how is the stack. Making numbers comparable is not a scoring formula, it is the identifier backbone that lets a play on one service and a play on another point to the same recording. Comparing across the fandom spiral is not a universal currency, it is the discipline of keeping paid and unpaid signals in separate columns joined by a shared identity key. And reliable attribution is not a smarter last-click model, it is running experiments. Each section below takes one layer, grounds it in the research, and says plainly what it asks you to do.

2. The backbone: identifiers and metadata

Everything above depends on being able to say, without ambiguity, *which recording, which song, and which person* a data point is about. The good news is the industry already has standards for exactly this. Three ISO codes form the base. The **ISRC** (International Standard Recording Code, ISO 3901) permanently identifies a specific sound recording or music video. IFPI administers the scheme, and the international ISRC database now contains over 150 million unique ISRCs registered with their associated recording and release data ([IFPI ISRC](#); [ISRC Handbook](#)). The **ISWC** (International Standard Musical Work Code, ISO 15707) identifies the underlying song, the composition, and CISAC's ISWC agency runs it for the collecting societies that pay songwriters and publishers ([CISAC identifiers](#); [ISO 15707](#)). The **ISNI** (International Standard Name Identifier, ISO 27729) tells apart the people involved, the writers, performers, producers, and publishers, and acts

as a bridge linking private rights databases to public discovery tools. Its value in music is precisely this disambiguation of people, which is what stops two artists with the same name drawing on each other's income ([ISNI](#); [ISO 27729](#)). The way one song maps to many recordings (ISWC to ISRC), tied back to the people by ISNI, is the join key the entire royalty system runs on.

Identifiers are useless if they cannot travel, and moving them through the supply chain is the job of the **DDEX** message standards. The Electronic Release Notification suite (ERN) is the format labels and distributors use to tell streaming services about a release, carrying the ISRC along with the deal terms, territories, rights, and takedowns, and ERN 4 adds a single consolidated list of every artist, writer, and label involved ([DDEX ERN](#)). Two companion standards matter for the argument that metadata is a growth lever and not just paperwork. The Media Enrichment and Description standard (MEAD) carries the richer stuff, lyrics, focus tracks, instrumentation, moods, that powers search and discovery, and the Recording Information Notification (RIN) captures credits at the moment of recording, in the studio, at the source ([DDEX MEAD](#); [DDEX standards](#)). DDEX's stated purpose for MEAD is itself the argument that metadata is a growth lever rather than paperwork: the standard exists to carry the sort of information that supports the marketing of music by DSPs, and DDEX expects it to expand as understanding of what serves that marketing improves. Figures are sometimes quoted for how much MEAD enrichment lifts plays and reduces skip rates; they are not stated on DDEX's own pages for the standard, so they are omitted here. Alongside the closed industry standards, the open **MusicBrainz** encyclopedia issues an MBID, a 36-character universally unique identifier permanently assigned to each entity in its database, covering artists, release groups, releases, recordings, works and labels, and offering a shared identifier layer as an alternative to locked silos ([MusicBrainz](#); [MBID](#)).

Get this layer wrong and the cost shows up as money that never reaches the artist. The US Mechanical Licensing Collective reports over \$4 billion in total streaming royalties distributed ([The MLC](#)), having passed \$3 billion by its annual membership meeting in October 2025. It also holds a pool it calls unclaimed accrued royalties: money it has received but has not been able to match or distribute despite its efforts, once the statutory minimum holding period expires. Under the Music Modernization Act the MLC is congressionally mandated to hand that pool onward, and it intends to begin doing so in 2027 through the market share distribution process, starting with blanket royalties from 2021 usage ([MLC unclaimed accrued royalties](#)). Market share is the operative phrase: unmatched money does not wait indefinitely for its owner, it is eventually shared out in proportion to catalogue size, which routes it to the largest rights-holders rather than the creators who earned it ([Revelator](#)). Estimates of the yearly black box vary a lot: the Ivors Academy in the UK has estimated it at £500m a year for streaming alone, while Billboard in the US put it at "at most \$250 million". The estimates are not reconcilable, but everyone agrees on the cause: a piece of music and its metadata travel down separate roads and fail to meet up ([Independent Music Insider](#)). The fix is to capture complete, standard metadata at the moment of creation instead of trying to reconstruct it later, which is the whole point of the industry's **Credits Due** campaign led by Björn Ulvaeus and the Ivors Academy ([Credits Due analysis](#)), and the job of credit-capture tools aimed at the studio, such as Sound Credit, which generates ISNI and ISRC identifiers directly ([Sound Credit](#)). The move for this layer is unglamorous and foundational: treat assigning identifiers and capturing credits as a contractual duty at the session, not a clean-up task for later.

3. From "looks related" to "actually caused it"

Part 1's second failure mode was reading a streaming spike that happens to coincide with a marketing push as though the push caused it. More data does not fix this. Only experiments do, and the evidence for that is some of the strongest in all of marketing research. The foundational result is [Lewis and Rao \(2015\)](#), who looked at twenty-five large field experiments with major US retailers and brokerages, most reaching millions of customers, and showed that measuring the returns to advertising is genuinely difficult: the median confidence interval on return on investment came out over 100 percentage points wide, because individual-level sales are very volatile relative to the per capita cost of the advertising, so an informative experiment can easily require more than 10 million person-weeks. If a controlled experiment struggles, then attribution with no experiment at all is hopeless. [Gordon and colleagues \(2019\)](#) made it concrete by pitting the usual observational methods against large randomised advertising experiments at Facebook, finding that observational methods often fail to accurately recover the treatment effects the experiments generated. This is not a tuning problem you can fix with better software. It is built into the situation.

The specific shortcuts the industry reaches for are the ones the research most directly discredits. [Berman \(2018\)](#) shows formally that the popular last-touch method overincentivises ad exposures and often results in lower advertiser profits, with the Shapley value performing better on that measure. His result also contains a counterintuitive finding worth flagging: the parties benefiting most from advertisers using last-touch are the popular publishers and those appearing early in the conversion funnel, so the distortion is not simply that the final click hogs the credit. The fancier multi-touch models meant to improve on last-click, whose lineage runs back to [Shao and Li \(2011\)](#), inherit the same flaw, because they still learn from what happened to occur together. The cautionary tale every music marketer should know is [Blake, Nosko and Tadelis \(2015\)](#): eBay's large-scale field experiments found that brand-keyword paid search, which the attribution reports had credited with real value, had no measurable short-term benefit, and that for non-brand keywords the frequent users whose purchasing was not influenced by ads accounted for most of the spend, making average returns negative. Because search clicks and purchase intent are correlated, returns from paid search came out a fraction of the non-experimental estimates. The music version is immediate. A playlist placement or an ad burst can look responsible for streams a fan was always going to generate.

The cure is a discipline called incrementality, and it has a mature toolkit. The menu of methods, controlled experiments, before-and-after comparisons, and the like, is written up for practitioners by [Varian \(2016\)](#) and in technical depth by [Yao and colleagues \(2021\)](#). When you cannot split an audience into a treated half and an untreated half, which is common in music because you cannot cut an artist's fanbase in two, the practical tool is the geo experiment, and [Chen and Au \(2022\)](#) give a robust way to measure return on ad spend from paired-region tests. [Johnson \(2023\)](#) offers a hands-on guide to running and reading field experiments in online display advertising, a medium he describes as hostile to experimentation because ad effects are tiny and the experimenter has limited control over exposure. And crucially, this has already been done for music: [Aguilar and Waldfogel \(2018\)](#) used Spotify playlist placements to estimate the true streaming value of promotion, rather than naively crediting a playlist with every stream that followed, which is the template for honest attribution. The move for this layer is to build a standing experiment

capability: default to incrementality and geo tests for marketing decisions, and treat any attribution number that did not come from an experiment as a hunch, not a measurement.

4. The fan graph you own

Part 1 established that the deepest, most valuable stages of the fandom spiral are mostly unpaid, and therefore invisible to streaming metrics, and that the superfan had become the industry's central growth story. The strategic consequence is that the industry has to build a *fan graph* it actually owns, a first-party record of who its fans are and what they do, because the platforms that handle most fan contact do not hand over the underlying data. The commercial case is now spelled out in numbers: Goldman Sachs identifies superfan monetisation as a significant opportunity, estimating a potential annual revenue uplift of \$4.3 billion based on 2026 projections, on the assumption that 20% of paid streaming subscribers can be defined as superfans who would spend twice as much as average subscribers, itself based on Luminate's finding that 20% of US music listeners count as superfans, and it pointedly describes today's flat-fee streaming as failing to segment by engagement "despite the wide availability of data" ([Goldman Sachs via MBW](#)). That last phrase is the whole problem in one line. The data exists, but nobody owns it, joins it, or segments it. That is a deluge, not a strategy.

The economics of direct-to-fan platforms show why owning the relationship pays. Comparative figures put Bandcamp's artist share at roughly 85 to 90 cents in the dollar, with Bandcamp taking 15% on digital and 10% on physical plus payment processing of about 4 to 7%, while Patreon takes a flat 10% of pledges for pages published after 4 August 2025 (earlier pages grandfathered on legacy tiers) plus processing. One operational finding is worth more than the fee comparison: artists who concentrate their paid superfan layer on a single platform retain 40 to 60% more recurring revenue at twelve months than artists fragmenting across three or more platforms ([D2F platform economics](#)). The theory of why fans will fund creators directly comes from the crowdfunding research: [Mollick \(2014\)](#) works out what makes a campaign succeed, [Agrawal, Catalini and Goldfarb \(2014\)](#) give the economic logic of cutting out the middleman between creator and backer, and [Colombo and colleagues \(2014\)](#) show that early, well-connected backers, the superfans, are the ones who get a campaign moving. For designing the tiered offers themselves, limited editions, memberships, price menus, the model of pricing and product decisions in reward crowdfunding by [Hu, Li and Shi \(2015\)](#) applies directly.

Building on data you own is not just commercially smart, it is becoming the only sustainable option as third-party data gets deprecated and regulated. [Martin and Murphy \(2020\)](#) review the privacy shifts pushing everyone toward first-party, consented data. The move for this layer applies Part 1's separate-columns principle to identity: capture the fan relationships you own (email, memberships, community, direct sales) as the spine of the fan graph, tie them to the identifier backbone from Section 2, and measure the unpaid stages of the spiral in their own column rather than forcing them into a stream-equivalent that erases what they meant.

5. Sharing data without spilling secrets

A fan graph and a real measurement capability both hit the same wall. The most valuable analyses need data held by different companies, a label's release data, a platform's listening data, a brand's

campaign data, but the data that would make the join precise is personal information that cannot legally or safely be pooled. The technology that squares this circle is the **data clean room**, a controlled space where two parties can compute a joint result without ever showing each other the raw, person-level records. The IAB Tech Lab has started standardising the practice, including an attribution protocol built on privacy-enhancing technologies (private set intersection, secure multiparty computation, trusted execution environments) so advertisers and publishers can measure outcomes without swapping identifiable data ([IAB Tech Lab](#)). Vendor versions add differential privacy, aggregation rules, and synthetic data ([Snowflake](#)). Honesty demands the caveat: these are not a finished answer. Privacy professionals note that clean-room vendors all describe similar powers without always explaining how identity is truly protected, and that the technology is necessary but not sufficient ([IAPP](#)).

The stronger privacy techniques come from outside advertising and carry over well. Federated analytics under multiparty homomorphic encryption has been shown to combine sensitive data across institutions with no raw data ever exposed, in precision medicine, by [Froelicher and colleagues \(2021\)](#), and the whole idea of training and analysing over data that never leaves its owner is surveyed definitively by [Kairouz and colleagues \(2021\)](#), which is a template for combining fan and usage data across labels and platforms without ever centralising personal information. The economic reason to bother with any of this is the value-versus-privacy trade-off laid out by [Acquisti, Taylor and Wagman \(2016\)](#). And the binding constraint is the law: the **GDPR** (Regulation (EU) 2016/679) sets the legal bases, consent rules, and cross-border limits governing any use of fan data ([GDPR primary text](#)), and its effects have spilled beyond Europe, increasing the market concentration of web technology vendors, as shown by [Peukert and colleagues \(2022\)](#) and synthesised by [Johnson \(2022\)](#). The move for this layer is to make privacy-preserving collaboration the default way to combine data across companies, clean rooms and federated computation rather than data dumps, and to treat governance and legal basis as design decisions, not compliance afterthoughts.

6. Governing what the AI learns from

The top layer governs the data that trains and drives the AI models, and it is where the events of Part 1, the Suno and Udio lawsuits, the flood of synthetic uploads, the streaming-fraud conviction, have turned voluntary good practice into emerging law. The starting point is the legal machinery for reserving your rights against having your work mined for training. [Flynn and colleagues \(2022\)](#) map the patchwork of global text-and-data-mining rules, and [Senftleben \(2025\)](#) analyses how the EU AI Act works with the EU's copyright opt-out to create a machine-readable "reserve your rights" regime that general-purpose AI providers have to respect. The ethical case that consent is owed at all is documented directly: [Jiang and colleagues \(2023\)](#) record the harm to creators from having their work used to train models without permission or payment, and [Lucchi \(2023\)](#) examines the copyright exposure of models trained on protected work.

What makes consent real rather than aspirational is infrastructure a machine can actually check. The prototype closest to what music needs is **DECORAIT**, a decentralised opt-in and opt-out registry that records a creator's consent and ownership and makes it machine-checkable by tying it to content provenance and fingerprinting ([Balan et al., 2023](#)). Its successor, "Content ARCs," extends the idea to bind provenance to licensing and payment flows, sketching the licensing-

marketplace layer of the playbook ([Balan et al., 2025](#)). The provenance foundation these build on is **C2PA**, the Coalition for Content Provenance and Authenticity, which provides an open technical standard letting publishers, creators and consumers establish the origin and edits of digital content. The standard is called Content Credentials, and C2PA's own analogy is a nutrition label: a record of the content's history that anyone can inspect at any time. Technically it is a cryptographically bound structure carrying a signature alongside details of the creator, the device and any edits applied, so a verifier can establish that the record is well formed, untampered and signed by a trusted party ([C2PA; spec](#)), bearing in mind the important caveat that C2PA certifies a content's history, not whether it is true. On the signal side, [AudioSeal](#) is the first audio watermarking technique designed for localised detection of AI-generated speech, achieving watermark detection down to the sample level. It is built for speech and voice cloning rather than music, so applying it to recorded music is an extrapolation rather than a demonstrated result, but it shows that signal-level marking can complement the manifest-based approach. That provenance labelling actually changes what people believe, and is therefore worth deploying, is shown by [Feng and colleagues \(2023\)](#), whose online experiment with 595 participants in the US and UK found that provenance indicators shift users' trust and accuracy perceptions.

Two developments show the market and the regulator arriving at the same destination. On the market side, the non-profit **Fairly Trained** certifies AI models that train only on licensed, consented, public-domain, or owned data, pointedly withholding its badge from models that lean on a fair-use defence, and it launched with nine already-certified generative AI companies spanning image, music and singing-voice generation, with Universal Music Group among the organisations expressing support ([Fairly Trained via Music Ally](#)). On the regulatory side, the EU AI Act's opt-out duty on general-purpose AI providers is set to be backed by a central, EUIPO-run rights-reservation registry, in effect a legally enforced "do-not-train" registry ([ECIJA analysis](#)). The move for this layer is to make consent and provenance machine-readable and attached at the source: reserve your rights in a registry, bind provenance credentials and watermarks to recordings, and prefer, and where possible certify, models trained on consented data.

7. The playbook in one page

The five layers add up to a single working discipline, and it helps to state them as commitments rather than aspirations.

1. **Fix identity first.** Assign ISRC, ISWC, and ISNI, and capture DDEX-standard metadata and credits at the moment of creation, by contract, because every capability above and every unpaid royalty below traces back to this layer.
2. **Own the fan graph.** Build a first-party record of fan relationships tied to those identifiers, and measure the unpaid stages of the fandom spiral in their own column instead of collapsing them into stream-equivalents.
3. **Measure cause, not coincidence.** Stand up an experiment capability and default to incrementality and geo tests, and treat any attribution number that did not come from an experiment as a hunch.
4. **Collaborate privately.** Make privacy-preserving architectures, clean rooms and federated computation, the default for combining data across companies, with legal basis and governance built in from the start.

5. Govern the model's inputs. Attach machine-readable consent and provenance to recordings, reserve rights in registries, and prefer models trained on consented data.

Each layer depends on the one below it. Skipping a layer does not save time, it just pushes the failure downstream to where it costs more to fix.

This is what the title of the original talk promised, and what the two years since have made urgent. Part 1 showed that abundance is not insight, and that AI has raised the cost of getting the data wrong from a bad marketing call to a machine making such calls at scale. Part 2's answer is that the response is neither more data nor a better dashboard, but a stack: identity at the base, an owned fan graph and honest measurement in the middle, private collaboration and machine-readable consent at the top. The industry already has every layer's tools, from ISO identifiers and DDEX standards, to two decades of experiment methodology, to direct-to-fan platforms, to privacy technology, to provenance and consent infrastructure that now carries legal weight. The move from deluge to strategy is no longer a question of capability. It is a question of deciding to build the stack, from the bottom up, before handing the decisions to the machines that will run on whatever it produces.

A companion bibliography of all verified sources for this part, organised by pillar with resolvable DOIs and dated URLs, accompanies this paper. See also Part 1 ("The Data Deluge") and its bibliography.